CPTG Native-Phase Jeans and Structural Validation

XMM-VID1-2075

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Abstract

Curvature Polarization Transport Gravity (CPTG) is tested here on XMM-VID1-2075, an exceptionally massive quiescent slow rotator observed at spectroscopic redshift $z = 3.44927$. JWST/NIRSpec measurements give an effective-radius stellar velocity dispersion $\sigma_{\star,R_e} = 387 \pm 22\,\mathrm{km\,s^{-1}}$, while JWST/NIRCam imaging gives a concentrated single-Sérsic profile with source-reported $R_e = 1.95 \pm 0.15\,\mathrm{kpc}$ and $n = 3.97 \pm 0.06$. Distance-dependent source quantities are re-expressed on the symmetric CPTG acoustic/luminosity midpoint distance coordinate before the bound-system solve; this changes the physical scale by only 0.997%. Redshift first selects CPTG native phase $N_{\pi,\mathrm{obs}} = -1.49274$ and native clock coordinate $t_{\pi,\mathrm{obs}} = 1.51879\,\mathrm{Gyr}$; that selected epoch is then treated as the local present of the bound-system reduction. The declared one-radius pressure-support estimator gives $a_\star = 2.45127 \times 10^{-9}\,\mathrm{m\,s^{-2}}$. An exact spherical Abel deprojection of the converted Sérsic source is normalized to the converted stellar mass and used to solve the stellar-only and CPTG radial fields before solving and projecting the Jeans equation. The isotropic stellar-only model gives $\sigma_{\mathrm{ap}} = 346.48\,\mathrm{km\,s^{-1}}$, whereas the CPTG field gives $394.16\,\mathrm{km\,s^{-1}}$, only 0.33σ from the observed value. A controlled $\beta_J = 11/50$ reconstruction gives $401.34\,\mathrm{km\,s^{-1}}$, also within 1σ. The same solved field gives Structural Mode $N_{\mathrm{sph}} = 3.04935$ and $N_{\mathrm{obl}} = 3.06602$. The calculation therefore supplies native-phase placement, baryon-source Jeans closure, and a post-solution structural classification without inserting a dark-halo gravitational term. Because the observed dispersion also enters the declared $a_\star$ estimator, the Jeans comparison is a shared-input closure test rather than a held-out prediction.

1 Introduction: Why XMM-VID1-2075 Matters

XMM-VID1-2075 stands out because several extreme properties are measured in the same early galaxy. It contains $3.3^{+0.1}_{-0.3} \times 10^{11} M_\odot$ in stars, has already become quiescent, and packs that stellar population into a half-light scale of only about 2 kpc [1]. JWST/NIRSpec integral-field spectroscopy further shows that the system is supported primarily by stellar velocity dispersion rather than by ordered rotation: the luminosity-weighted dispersion inside the effective radius is $\sigma_{\star,R_e} = 387 \pm 22\,\mathrm{km\,s^{-1}}$, the central bins approach $500\,\mathrm{km\,s^{-1}}$, and every measured stellar velocity offset satisfies $|V_\star| < 100\,\mathrm{km\,s^{-1}}$ [1].

The low angular momentum is especially unusual at this redshift. The version-of-record analysis reports the resolved stellar spin parameter

$$\lambda_{R_e} = 0.123^{+0.073}_{-0.023}, \tag{1}$$

placing XMM-VID1-2075 securely in the slow-rotator regime [1]. Deep imaging also reveals low-surface-brightness asymmetries consistent with past merger activity. The observations do not uniquely identify the formation path: Forrest et al. retain major mergers, accumulated minor mergers, and transformation associated with isotropic gas inflow as distinguishable possibilities [1]. What is directly established is that a very massive, compact, quiescent galaxy had already reached a strongly dispersion-dominated state by the observed epoch.

This combination makes the galaxy useful for a gravitational test. Rotation-supported galaxies can be confronted with a gravity model through their circular-velocity curves. XMM-VID1-2075 instead requires a pressure-supported treatment: the gravitational field must be propagated through the stellar Jeans equation and compared with the measured velocity dispersion.

Curvature Polarization Transport Gravity (CPTG) treats the observed baryonic system as the source and computes a nonlinear curvature-polarization/transport response without adding a dark-halo mass component to the local field equation [7]. The object-dependent structural acceleration scale $a_\star$ sets the response scale of the bound system. After the field has been solved, CPTG can also ask a second question that is independent of the dispersion comparison: how is that solved radial response organized inside the galaxy? That post-solution organization is measured by Structural Mode N and translated, when useful, into a descriptive CSMI structural type [9].

The paper therefore proceeds in a fixed order. The redshift first selects the common CPTG native phase. The measured stellar structure is then reconstructed as a three-dimensional source, the local $a_\star$ reduction is declared, the CPTG field is solved at each radius, and that field is propagated through the Jeans equation. Only after the field solution is complete is Structural Mode N evaluated. This ordering keeps cosmological placement, local dynamical support, and structural classification logically separate.

2 Purpose, Test Definition, and Observed Inputs

The purpose of the calculation is to test whether one observed galaxy can be carried through the CPTG inference chain without importing a dark-halo gravitational source. The calculation is not a fit of every available galaxy property. Each observable has a specific role. Redshift fixes the geometric epoch; the stellar mass and light profile define the baryonic source; the effective-radius dispersion supplies the declared pressure-support estimate of $a_\star$; the solved field is then tested through the Jeans equation; and the retained radial field finally supplies Structural Mode N.

Operationally, the calculation follows [7, 8, 9]

$$\begin{gathered} z \longrightarrow N_\pi \longrightarrow \text{selected native epoch} \\ \Downarrow \\ \{M_\star, R_e, n, \sigma_\star\} \longrightarrow a_\star \longrightarrow g_{\rm CPTG}(r) \longrightarrow \begin{cases} \sigma_{\rm ap}(< R_e), \\ N. \end{cases} \end{gathered} \tag{2}$$

The symbols in Eq. (2) represent different layers of the theory. N_π is a cosmological phase coordinate selected by redshift. It is *not* Structural Mode N. The quantity $a_\star$ is the object-dependent structural acceleration scale of the galaxy at the selected local present. The function $g_{\rm CPTG}(r)$ is the solved radial gravitational field. The aperture quantity $\sigma_{\rm ap}(< R_e)$ is the observable pressure-support prediction obtained after the field has been passed through the Jeans equation. Structural Mode N is evaluated only afterward and measures the radial organization of the retained solution.

Redshift is therefore used only for cosmological placement. Once the observed phase is selected, that epoch is the local present for the bound-system reduction. No factor of $H(z)$ or $E(z)$ is used to evolve the intrinsic object-dependent $a_\star$.

Forrest et al. report several consistent redshift determinations and use the EMILES result $z_E = 3.44927$ for the kinematic analysis [1]. The present paper adopts that value. The published mass and physical-radius measurements are retained as the observational authority, but their distance-dependent units require one bookkeeping step before they enter the CPTG bound-system calculation. That conversion is described immediately after the source-input table; it changes only the reporting coordinate and does not alter the observed redshift, velocity dispersions, Sérsic index, ellipticity, or spin parameter.

Observed quantity	Adopted value	Role in test
Kinematic redshift	$z = 3.44927$	Common native phase
Stellar mass	$3.3^{+0.1}_{-0.3} \times 10^{11} M_\odot$	Stellar source normalization
Star-formation rate	$< 1\,M_\odot\,\mathrm{yr}^{-1}$	Quiescent-state context
Effective-radius dispersion	$387 \pm 22\,\mathrm{km\,s^{-1}}$	Pressure-support anchor and comparator
Ground-based dispersion	$379^{+85}_{-53}\,\mathrm{km\,s^{-1}}$	Independent scalar consistency check
Central stellar dispersion	$\sim 500\,\mathrm{km\,s^{-1}}$	Resolved core-gradient context
IFU half-light geometry	$R_{e,\mathrm{maj}} = 2.25 \pm 0.37\,\mathrm{kpc}$, $\epsilon = 0.11 \pm 0.03$	Kinematic aperture geometry
IFU circularized effective radius	$R_{e,\mathrm{circ}} = 2.00\,\mathrm{kpc}$	Jeans aperture and $a_\star$ estimator
NIRCam Sérsic structure	$R_e = 1.95 \pm 0.15\,\mathrm{kpc}$, $n = 3.97 \pm 0.06$	Radial stellar-source shape
NIRCam ellipticity	$\epsilon = 0.19 \pm 0.01$	Oblate-source geometry stress test
Spin parameter	$\lambda_{R_e} = 0.123^{+0.073}_{-0.023}$	Resolved slow-rotator classification

Table 1: Source-reported inputs used to define the XMM-VID1-2075 test. The IFU half-light ellipse defines the kinematic aperture, while the independently fitted NIRCam Sérsic profile supplies the radial stellar-source shape. Distance-dependent physical quantities are converted below before entering the CPTG bound-system solve.

2.1 Source-distance convention and CPTG conversion

The source papers report physical sizes and stellar masses rather than only angular measurements. Those quantities inherit an observational distance calibration. The MAGAZ3NE analysis that supplies the stellar-mass and ground-based dynamical context for XMM-VID1-2075 explicitly adopts flat-ΛCDM with $H_0 = 70\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, $\Omega_m = 0.3$, and $\Omega_\Lambda = 0.7$ [2]. The later JWST paper reports the same object at the same redshift and gives a physical IFU scale of about 730 pc per $0.1''$ spaxel, consistent with that source-distance convention [1]. These standard-cosmology quantities are used here only to identify the source reporting scale. They do not govern the CPTG native branch or its clock.

For a neutral conversion of source-reported physical units, this paper follows the same convention used in the recent CPTG high-redshift structural tests: it takes the arithmetic midpoint of the acoustic and luminosity *distance coordinates*. The midpoint is a conversion coordinate, not an additional cosmological branch. For $b \in \{\mathrm{ac}, \mathrm{lum}\}$,

$$D_{M,b}(z) = \frac{c}{H_{0,b}} \int_0^z \frac{dz'}{E_b(z')}, \qquad D_{A,b}(z) = \frac{D_{M,b}(z)}{1+z}, \tag{3}$$

and

$$D_{A,\mathrm{mid}}(z) = \frac{D_{A,\mathrm{ac}}(z) + D_{A,\mathrm{lum}}(z)}{2}. \tag{4}$$

At $z = 3.44927$,

$$D_A^{\Lambda 70} = 1518.1765\,\mathrm{Mpc}, \tag{5}$$

$$D_{A,\mathrm{ac}} = 1561.4422\,\mathrm{Mpc}, \tag{6}$$

$$D_{A,\mathrm{lum}} = 1444.6470\,\mathrm{Mpc}, \tag{7}$$

$$D_{A,\mathrm{mid}} = 1503.0446\,\mathrm{Mpc}. \tag{8}$$

Thus the distance scale applied to source-reported physical lengths is

$$s_D \equiv \frac{D_{A,\mathrm{mid}}}{D_A^{\Lambda 70}} = 0.99003286. \tag{9}$$

The correction is only -0.997% in length. At fixed observed flux, the leading distance dependence of the stellar-mass normalization is $M_\star \propto D_L^2$; at fixed redshift the luminosity- and angular-diameter-distance ratios are the same, so the corresponding leading mass factor is $s_D^2 = 0.98016506$. This is a source-unit conversion, not a new stellar-population fit.

Distance-dependent quantity	Source-reported value	CPTG midpoint value
Stellar mass	$3.3^{+0.1}_{-0.3} \times 10^{11} M_\odot$	$3.235^{+0.098}_{-0.294} \times 10^{11} M_\odot$
IFU semimajor R_e	$2.25 \pm 0.37\,\mathrm{kpc}$	$2.228 \pm 0.366\,\mathrm{kpc}$
IFU circularized R_e	$2.00\,\mathrm{kpc}$	$1.98007\,\mathrm{kpc}$
NIRCam Sérsic R_e	$1.95 \pm 0.15\,\mathrm{kpc}$	$1.93056 \pm 0.14850\,\mathrm{kpc}$

Table 2: Distance-dependent source quantities before and after the neutral CPTG acoustic/luminosity midpoint conversion. Dimensionless morphology and velocity measurements are unchanged.

With the observational roles and source-unit conversion fixed, the first theoretical step is to place the galaxy at one CPTG geometric epoch before any bound-system quantity is evaluated.

3 Native Phase and Bound-System Epoch

In CPTG, a high-redshift galaxy is not first assigned a standard-cosmology age and then evolved locally from that age. Its observed redshift first selects a point on the native geometric branch. That common phase can then be expressed through native, acoustic, and luminosity clock coordinates, but the three clocks refer to the same selected geometric event [7, 8]. Once that event is selected, it becomes the local present for the galaxy calculation.

The notation deserves an explicit distinction. The cosmological quantity

$$N_\pi = \ln a \tag{10}$$

is a branch-phase coordinate. It is unrelated to the galaxy Structural Mode N introduced later in the paper. The two quantities answer different questions and are never substituted for one another.

For the branch construction, define

$$\mathcal{B}_\pi = 1 - \frac{1}{\pi}, \qquad \mathcal{T}_\pi = \frac{1}{\pi}, \qquad p_{\rm ac} = 3 - \frac{\pi}{100}, \tag{11}$$

$$\Omega_{m,\rm ac} = \frac{p_{\rm ac}}{3\pi}, \qquad \Omega_{\Lambda,\rm ac} = 1 - \Omega_{m,\rm ac}. \tag{12}$$

The corresponding comparison coordinates are

$$E_\pi(a)^2 = \mathcal{B}_\pi + \mathcal{T}_\pi a^{-\pi}, \tag{13}$$

$$E_{\rm ac}(a)^2 = \mathcal{B}_\pi + \mathcal{T}_\pi a^{-p_{\rm ac}}, \tag{14}$$

$$E_{\rm lum}(a)^2 = \Omega_{\Lambda,\rm ac} + \Omega_{m,\rm ac} a^{-p_{\rm ac}}, \tag{15}$$

with present normalizations

$$H_0^{(\pi)} = 69.4162507897, \quad H_{0,\rm ac} = 67.4777967351, \quad H_{0,\rm lum} = 73.17 \tag{16}$$

in $\mathrm{km\,s^{-1}\,Mpc^{-1}}$.

At $z = 3.44927$,

$$a = \frac{1}{1+z} = 0.2247559712, \qquad N_{\pi,\rm obs} = \ln a = -1.492740038. \tag{17}$$

For a branch of the form $E^2 = B + Ta^{-q}$, the integrated clock coordinate is

$$t_b(a) = \frac{2}{qH_{0,b}\sqrt{B}} \sinh^{-1}\left(\sqrt{\frac{B}{T}}\, a^{q/2}\right). \tag{18}$$

The common observed phase therefore has the branch-coordinate readouts

Coordinate	$E_b(z)$	$t_b(z)$ (Gyr)
Native π branch	5.942745	1.518785
Acoustic projection	5.237680	1.879594
Luminosity projection	5.211188	1.742407

Table 3: Branch-coordinate readouts at the common observed phase of XMM-VID1-2075. The different clock values are projections of one native phase.

The standard-cosmology statement that the galaxy is observed when the Universe is about 1.8 Gyr old is retained only as source context. CPTG uses the branch coordinates above for cosmological placement. No branch clock subsequently rescales the galaxy's measured $a_\star$ or replaces the local bound-system calculation.

Having fixed the epoch, the next question is physical rather than chronological: what gravitational accounting is required to support such a massive, already dispersion-dominated galaxy?

4 Standard ΛCDM Formation Challenge and CPTG Contrast

The contrast with the standard dark-matter framework has two logically separate parts. The first is a formation-history problem: how did an object this massive become compact, quiescent, and nearly devoid of ordered stellar rotation so early? The second is a present-state gravitational problem: given the stellar system that is actually observed, what gravitational field is required to support its measured dispersion? Keeping those questions separate prevents formation astrophysics from being confused with the local gravity test.

In the standard ΛCDM picture, massive galaxies form and evolve inside dark-matter haloes. XMM-VID1-2075 is challenging because an extremely massive stellar system is already quiescent and has already lost nearly all ordered stellar rotation by $z \simeq 3.45$. Slow rotators are expected to be much less common at such early epochs, while the canonical low-redshift route invokes merger-driven angular-momentum loss and structural transformation [4, 1].

The observations do not uniquely identify one formation path. Forrest et al. show that the low-surface-brightness asymmetries are consistent with merger activity, but they explicitly retain major mergers, cumulative minor mergers, and isotropic gas-inflow-driven transformation as competing possibilities [1]. Earlier work on this same galaxy also identified the rapid buildup and quenching of its stellar mass as difficult for then-current theoretical models [3]. The relevant standard-model challenge is therefore the complete halo-plus-baryon history required to assemble, quench, and dynamically restructure such a system so early, not a claim that one particular head-on merger has been observed.

CPTG addresses a different part of the problem. It does not replace the baryonic microphysics of star formation, mergers, gas inflow, black-hole growth, or feedback. It changes the gravitational accounting of the measured bound system. In the calculation below, the measured stellar source is inserted into the CPTG field equation and no dark-halo gravitational source is added.

Question	Standard dark-matter framework	CPTG calculation in this paper
Galaxy gravitational source	Baryons embedded in a gravitating dark-matter halo	Measured stellar source plus curvature-polarization/transport response
Stellar-only Jeans field	Incomplete baseline; a halo can supply additional gravitational support	Explicit comparison baseline before CPTG response
Dark-halo term in local solve	Available as an additional gravitating component	Not inserted
Early slow-rotator formation	Halo assembly plus merger/inflow and feedback histories	Formation microphysics left to astrophysics; local gravitational state tested separately
Additional structural output	No CPTG Structural Mode analogue	Continuous Structural Mode N from the solved field

Table 4: Conceptual separation of the standard dark-matter interpretation and the CPTG bound-system calculation. The stellar-only Jeans solution is not itself a full ΛCDM model; it is the common baryonic baseline against which the CPTG geometric response is evaluated.

The local comparison can now begin from quantities that are actually measured: the galaxy's stellar mass distribution and its pressure-supported kinematics.

5 Stellar Source and Local Structural Acceleration Scale

A Jeans calculation requires a three-dimensional tracer density, whereas the telescope measures a two-dimensional light distribution on the sky. The NIRCam Sérsic fit therefore provides the starting geometry, and the paper adopts a constant stellar mass-to-light ratio so that the observed light profile can be used as the shape of the stellar mass profile. This is a declared modeling reduction, not an additional fitted freedom.

The resolved NIRCam fit supplies

$$\Sigma_\star(R) \propto \exp\left[-b_n\left(\left(\frac{R}{R_e}\right)^{1/n} - 1\right)\right], \tag{19}$$

with converted $R_e = 1.93056\,\mathrm{kpc}$ and $n = 3.97$. The spherical-equivalent three-dimensional density is obtained by Abel inversion,

$$\rho_\star(r) = -\frac{1}{\pi}\int_r^\infty \frac{d\Sigma_\star/dR}{\sqrt{R^2 - r^2}}\,dR, \tag{20}$$

and is normalized to the distance-converted stellar mass $M_\star = 3.23454 \times 10^{11} M_\odot$. The resulting stellar-source acceleration field is

$$g_\star(r) = \frac{GM_\star(< r)}{r^2}. \tag{21}$$

CPTG also requires the object's structural acceleration scale $a_\star$. This is not a universal constant. It is an object-dependent acceleration scale associated with the gravitating system. For this pressure-supported one-radius reduction, the observed dispersion and circularized effective radius provide the declared estimator

$$a_\star \simeq \frac{\sigma^2_{\star,R_e}}{R_{e,\mathrm{circ}}}. \tag{22}$$

Using $\sigma_{\star,R_e} = 387\,\mathrm{km\,s^{-1}}$ and the converted $R_{e,\mathrm{circ}} = 1.98007\,\mathrm{kpc}$ gives

$$a_\star = 2.45127 \times 10^{-9}\,\mathrm{m\,s^{-2}}. \tag{23}$$

The quoted dispersion uncertainty alone gives approximately

$$2.18 \times 10^{-9} \lesssim a_\star \lesssim 2.74 \times 10^{-9}\,\mathrm{m\,s^{-2}}. \tag{24}$$

This estimator is the pressure-support bridge used in the present calculation. It does not assert that $a_\star$ is universally equal to σ^2/R in every CPTG system, and it is not evolved by the cosmological branch clocks. At this point the two local ingredients are fixed: $g_\star(r)$ describes the measured stellar source, while $a_\star$ sets the response scale of this galaxy. The next step is to solve the CPTG field generated from them.

6 Radial CPTG Field Response

The stellar field $g_\star(r)$ is the baryonic source field obtained from the measured mass profile. CPTG does not replace that source. Instead, it solves a nonlinear curvature-polarization/transport response around it. The response depends on the local stellar field, the object-dependent scale $a_\star$, and the geometric transport term $g_{\rm geom}(r)$ [7].

The bound-system field is solved independently at every radius from

$$g(r) = g_\star(r) + \frac{\sqrt{a_\star g_\star(r)} + g_{\rm geom}(r)}{\left[1 + (g(r)/a_\star)^{123/50}\right]^{11/50}}, \tag{25}$$

where

$$g_{\rm geom}(r) = \frac{1}{2} a_\star \frac{(r/R_c)^{7/5}}{1 + (r/R_c)^{12/5}}, \qquad R_c = \frac{48\pi c^2}{a_\star}. \tag{26}$$

The scale R_c is derived from $a_\star$ and controls the geometric transport continuation. For Eq. (23),

$$R_c = 179.180\,\mathrm{Gpc}. \tag{27}$$

At the IFU circularized effective radius, the exact Sérsic deprojection gives

$$g_\star(1.98007\,\mathrm{kpc}) = 4.85370 \times 10^{-9}\,\mathrm{m\,s^{-2}}, \tag{28}$$

while the solved CPTG field is

$$g_{\rm CPTG}(1.98007\,\mathrm{kpc}) = 6.80487 \times 10^{-9}\,\mathrm{m\,s^{-2}}. \tag{29}$$

The local ratio is therefore

$$\frac{g_{\rm CPTG}}{g_\star} = 1.40200, \tag{30}$$

so at R_e the solved CPTG field is about 40.2% larger than the stellar-source field alone.

The formal transport remnant at this radius is only

$$g_{\rm geom}(1.98007\,\mathrm{kpc}) = 8.89 \times 10^{-21}\,\mathrm{m\,s^{-2}}, \tag{31}$$

so the local enhancement is overwhelmingly polarization-dominated in this regime.

A single value at R_e is not enough to predict a stellar velocity dispersion, because the Jeans equation responds to the gravitational field over the full range of stellar orbits contributing to the aperture. The CPTG-to-stellar field ratio also changes with radius. For that reason, the calculation does not rescale one effective-radius dispersion by a local field factor. It passes the complete solved radial field into the Jeans equation.

7 Direct Dispersion-Supported Jeans Reconstruction

For a rotating disk, a gravity model can be tested by asking whether its radial acceleration reproduces the observed circular velocity. For a dispersion-supported galaxy, the analogous bridge is the Jeans equation. It connects the gravitational field to the random stellar motions required to support the tracer population against that field. The result is then projected along the line of sight and averaged over the same aperture used by the observation.

For a spherical stellar tracer in steady state,

$$\frac{d(\rho_\star \sigma_r^2)}{dr} + \frac{2\beta_J(r)}{r}\rho_\star \sigma_r^2 = -\rho_\star g(r). \tag{32}$$

The anisotropy parameter β_J describes how the stellar velocity dispersion is divided between radial and tangential orbital directions. $\beta_J = 0$ is isotropic; positive values give progressively more radial orbital support. Defining

$$F(r) = \exp\left[\int^r \frac{2\beta_J(s)}{s}\, ds\right], \tag{33}$$

the physical outer-boundary solution is

$$\sigma_r^2(r) = \frac{1}{\rho_\star(r)F(r)} \int_r^\infty \rho_\star(s)F(s)g(s)\, ds. \tag{34}$$

The line-of-sight second moment is projected as [5, 6]

$$\Sigma_\star(R)\sigma_{\rm los}^2(R) = 2\int_R^\infty \left(1 - \beta_J(r)\frac{R^2}{r^2}\right) \frac{\rho_\star(r)\sigma_r^2(r) r\, dr}{\sqrt{r^2 - R^2}}, \tag{35}$$

and the observable comparison is the luminosity-weighted aperture moment inside the IFU half-light radius,

$$\sigma_{\rm ap}^2(< R_e) = \frac{\displaystyle\int_0^{R_e} \Sigma_\star(R)\sigma_{\rm los}^2(R)\, 2\pi R\, dR}{\displaystyle\int_0^{R_e} \Sigma_\star(R)\, 2\pi R\, dR}. \tag{36}$$

Two anisotropy cases are retained so that the result does not depend on a single orbital assumption. The neutral primary comparison is isotropic,

$$\beta_J = 0. \tag{37}$$

A second controlled CPTG branch sensitivity uses

$$\beta_J = \frac{11}{50} = 0.22. \tag{38}$$

The latter is not treated as an observed property of XMM-VID1-2075. It is a fixed theory sensitivity test. No outer anisotropy transition is fitted to the galaxy in this paper.

The same tracer density and stellar mass normalization are solved twice: first with $g(r) = g_\star(r)$ and then with $g(r) = g_{\rm CPTG}(r)$. This isolates the change produced by the CPTG gravitational response while keeping the observed stellar source unchanged.

Jeans case	$\sigma_{\rm ap}$ (km s^{-1})	Residual (km s^{-1})	Observed offset
Observed	387 ± 22	—	—
Stellar only, $\beta_J = 0$	346.48	-40.52	-1.84σ
CPTG, $\beta_J = 0$	394.16	$+7.16$	$+0.33\sigma$
Stellar only, $\beta_J = 11/50$	350.47	-36.53	-1.66σ
CPTG, $\beta_J = 11/50$	401.34	$+14.34$	$+0.65\sigma$

Table 5: Direct spherical Jeans reconstruction using the full radial stellar-only and CPTG fields. The stellar-only rows are a baryonic Newtonian baseline, not a complete ΛCDM halo model.

For the neutral isotropic reconstruction, the absolute central dispersion residual is reduced by 82.3% when the CPTG field replaces the stellar-only field. For the fixed $\beta_J = 11/50$ sensitivity, the reduction is 60.7%. Both CPTG reconstructions lie inside the quoted 1σ observational interval. The independent ground-based measurement 379^{+85}_{-53} km s^{-1} is also consistent with both values.

The observed dispersion enters Eq. (23), so these offsets are not an independent significance test of a held-out prediction. The nontrivial closure question is narrower: after the observed pressure scale has fixed $a_\star$, does the independently specified CPTG radial field, combined with the measured stellar source, return a self-consistent aperture-supported state? It does for both anisotropy cases above.

The Jeans result answers how much dynamical support the solved field provides. CPTG Structural Mode asks a different question of that same retained field: where, radially, is its structural variation organized?

8 Structural Mode *N* and CSMI

The Jeans calculation tests dynamical support. Structural Mode N tests radial organization. It is a dimensionless, post-solution diagnostic evaluated only after the CPTG field has been solved. It is not a measure of total gravitational strength, not another fit parameter, not a visual morphology, and not the cosmological phase coordinate N_π. In practical terms, N summarizes where the solved radial gravitational response is most strongly changing relative to the adopted outer structural radius [9].

The construction begins by combining the field amplitude and its radial gradient into

$$C_g(r) = g(r)^2 + \kappa_{\rm struct}^2 \left(\frac{dg}{dr}\right)^2, \qquad \kappa_{\rm struct} = 1\,{\rm kpc}. \tag{39}$$

The derivative

$$w(r) = \left|\frac{dC_g}{dr}\right| \tag{40}$$

then acts as a structural weight: radii at which the solved field changes most strongly contribute most heavily to the organization scale

$$\lambda = \frac{\int_{R/5}^{R} r\,w(r)\,dr}{\int_{R/5}^{R} w(r)\,dr}. \tag{41}$$

The final Structural Mode is

$$N = \frac{R}{\lambda}. \tag{42}$$

The interpretation follows directly from this ratio. If the structural weight is concentrated farther inward, λ is smaller relative to R and N is larger. If the dominant structural variation remains farther outward, λ is larger and N is smaller. Thus N measures organization of the solved field rather than simply repeating $a_\star$, mass, size, or field amplitude.

For galaxy work, the continuous mode can be translated into the current CSMI catalog layer. In the present CPTG workbench, CSMI is the Curvature Structure Mode Index: the numerical Structural Mode N is solved first, and only afterward is its interval mapped to a descriptive CSMI Type. The inference direction is

$$\text{observed source} \longrightarrow \text{solved CPTG field} \longrightarrow N \longrightarrow \text{CSMI Type}. \tag{43}$$

No Hubble type, galaxy name, or visual morphology is supplied to the calculation of N. The familiar CSMI names are therefore descriptive catalog labels for solved structural populations, not morphology inputs and not one-to-one replacements for conventional Hubble classes [9].

Structural Mode N	Current CSMI Type
$N \leq 1.45$	Dwarf Irregular
$1.45 < N \leq 1.75$	Magellanic Irregular
$1.75 < N \leq 1.95$	LSB Dwarf Disk
$1.95 < N \leq 2.15$	Transition Dwarf
$2.15 < N \leq 2.35$	LSB Spiral
$2.35 < N \leq 2.55$	Very Late Spiral
$2.55 < N \leq 2.80$	Late Spiral
$2.80 < N \leq 3.05$	Intermediate Spiral
$3.05 < N \leq 3.30$	Early Spiral
$3.30 < N \leq 3.55$	Bulged Spiral
$3.55 < N \leq 3.72$	Lenticular/Early Disk
$N > 3.72$	High-Mode Outlier

Table 6: Current CSMI catalog map associated with the validated galaxy Structural Mode implementation. The continuous value of N is the primary scientific result; the named type is assigned only afterward.

For the fitted Sérsic source, the registered outer structural radius is the projected 95-

percent-light radius,

$$R = R_{95} = 16.53532\,\mathrm{kpc}. \tag{44}$$

The spherical-equivalent reconstruction gives

$$\lambda_{\mathrm{sph}} = 5.42257\,\mathrm{kpc}, \qquad N_{\mathrm{sph}} = 3.04935. \tag{45}$$

This lies immediately below the current $N = 3.05$ Intermediate-Spiral/Early-Spiral CSMI boundary. Because the value is boundary-level, the continuous mode is more informative than the discrete name.

The NIRCam source is not perfectly spherical. Its measured ellipticity $\epsilon = 0.19 \pm 0.01$ corresponds to projected axis ratio $q_{\mathrm{proj}} = 0.81 \pm 0.01$. As a non-fitted source-geometry stress test, consider similar oblate density surfaces with

$$m^2 = R^2 + \frac{z^2}{q_{\mathrm{int}}^2}. \tag{46}$$

For the edge-on endpoint $q_{\mathrm{int}} = q_{\mathrm{proj}}$, the exact equatorial field of nested oblate homoeoids is [5]

$$g_{\star,\mathrm{obl}}(R) = \frac{4\pi G q_{\mathrm{int}}}{R} \int_0^R \frac{\rho_q(m) m^2\, dm}{\sqrt{R^2 - (1 - q_{\mathrm{int}}^2) m^2}}. \tag{47}$$

Solving the unchanged CPTG response with that source gives

$$\lambda_{\mathrm{obl}} = 5.39309\,\mathrm{kpc}, \qquad N_{\mathrm{obl}} = 3.06602. \tag{48}$$

The independently measured IFU half-light ellipticity $\epsilon = 0.11$ gives $q = 0.89$ and, inserted into the same geometry stress test, gives

$$N_{\mathrm{obl,IFU}} = 3.05878. \tag{49}$$

Both measured flattening choices therefore move the continuous mode only slightly upward across the current $N = 3.05$ catalog boundary.

Structural reconstruction	N	CSMI readout
Spherical-equivalent source	3.04935	Intermediate Spiral, boundary-level
IFU flattening, $q = 0.89$	3.05878	Early Spiral
NIRCam flattening, $q = 0.81$	3.06602	Early Spiral
NIRCam $R_e = 2.07907\,\mathrm{kpc}$ sensitivity	3.04410	Intermediate Spiral
NIRCam $R_e = 1.78206\,\mathrm{kpc}$ sensitivity	3.09446	Early Spiral

Table 7: Structural Mode results and the dominant effective-radius sensitivity. The CSMI names classify solved CPTG structural organization; they are not visual morphology labels. XMM-VID1-2075 remains observationally an elliptical slow rotator.

One-at-a-time variations of the measured NIRCam Sérsic index, stellar mass, stellar dispersion, and ellipticity keep the oblate central result within approximately 3.054–3.080.

The quoted NIRCam effective-radius range broadens that interval to approximately 3.044–3.094. The robust result is therefore the continuous placement of XMM-VID1-2075 at the Intermediate/Early structural-mode transition. The fact that the current CSMI names use spiral terminology does not turn the observed galaxy into a spiral; it means that the solved CPTG field falls in a structural interval originally named from the validated galaxy-mode population.

The Jeans and Structural Mode calculations therefore extract two different readouts from the same field: one measures the amount of pressure support, while the other measures how the radial response is organized. The following section combines those results and states the remaining observational limitations.

9 Results, Interpretation, and Limitations

The calculation now supports three separate statements, each tied to a different stage of the inference chain.

First, the observed redshift places XMM-VID1-2075 at native phase $N_{\pi,\mathrm{obs}} = -1.492740038$ and native time coordinate $t_{\pi,\mathrm{obs}} = 1.518785\,\mathrm{Gyr}$. These are cosmological placement coordinates only. They do not define an $a_\star(t)$ law for the bound system, and they are not the galaxy Structural Mode N.

Second, the direct radial Jeans reconstruction closes the observed effective-radius pressure support without a dark-halo term. For the neutral isotropic case, the stellar-only source gives $346.48\,\mathrm{km\,s^{-1}}$, while the same source under the CPTG field gives $394.16\,\mathrm{km\,s^{-1}}$ against the measured $387 \pm 22\,\mathrm{km\,s^{-1}}$. The $\beta_J = 11/50$ sensitivity gives $350.47\,\mathrm{km\,s^{-1}}$ and $401.34\,\mathrm{km\,s^{-1}}$, respectively. The result is therefore a successful effective-radius closure of the observed pressure-supported state under the declared reduction.

Third, the same solved field supplies an independent radial-organization measurement. Structural Mode is $N_{\mathrm{sph}} = 3.04935$ in the spherical-equivalent reconstruction and $N_{\mathrm{obl}} = 3.06602$ under the measured NIRCam flattening. The corresponding CSMI readout lies at the current Intermediate/Early boundary. This output is conceptually distinct from the Jeans dispersion: σ_{ap} measures projected dynamical support, while N measures where the solved field's structural variation is concentrated.

The dark-matter contrast should remain precise. The stellar-only rows in Table 5 are not numerical predictions of a complete ΛCDM halo model. A conventional ΛCDM dynamical model can add a dark halo to the stellar source. The result established here is different: CPTG obtains an aperture-supported state consistent with the observation from the measured stellar source plus its curvature-polarization/transport response, without inserting a dark-halo gravitational term.

Several limitations define the current scope. The distance conversion uses the source paper's declared ΛCDM reporting convention and a leading D_L^2 rescaling of stellar mass rather than a fresh stellar-population refit; the correction is only about one percent in length and two percent in mass. The stellar deprojection assumes constant stellar M/L. The Jeans calculation is spherical even though the source has measurable flattening; the oblate calculation is presently used only for the Structural Mode geometry stress test, not for a full axisymmetric Jeans solution. The observed dispersion is used to estimate $a_\star$, so the dispersion comparison is a shared-input closure test rather than a fully independent prediction. The published source also contains resolved Voronoi-bin kinematics, but this paper does not fit those bins individually. The continuous Structural Mode calculation itself is independent of morphology, but applying the present CSMI names to a high-redshift elliptical is an out-of-sample use of a catalog layer validated primarily on the SPARC galaxy population; for this object the continuous N value therefore carries more authority than the descriptive CSMI name. Finally, the R_{95} Structural Mode calculation extrapolates the fitted Sérsic profile into the low-surface-brightness outer region, where the observed merger-related asymmetries make deeper structural measurements particularly valuable.

These limitations do not erase the present result; they identify the next level of falsifiability. A stronger test would use the resolved NIRSpec bins directly, allow axisymmetric orbital support, and recompute Structural Mode from the best available resolved stellar-mass profile rather than from a single-Sérsic continuation.

10 Conclusion

XMM-VID1-2075 is an unusually demanding early-galaxy test because its extreme stellar mass, compact structure, quiescence, and exceptionally low stellar spin are all measured in the same system. The observations establish a galaxy whose dynamics are dominated by dispersion rather than ordered rotation, so a rotation-curve reduction is inappropriate. The correct comparison is a pressure-supported Jeans reconstruction.

The CPTG calculation first places the observation at $N_{\pi,\mathrm{obs}} = -1.492740038$ on the native branch and then treats that selected epoch as the local present of the galaxy. At that epoch, the declared one-radius reduction gives $a_\star = 2.45127 \times 10^{-9}\,\mathrm{m\,s^{-2}}$. Using the distance-converted stellar mass and NIRCam Sérsic structure, the complete radial field is solved before the Jeans equation is projected into the observed aperture.

For the isotropic case, the stellar-only reconstruction gives $346.48\,\mathrm{km\,s^{-1}}$. CPTG gives $394.16\,\mathrm{km\,s^{-1}}$, compared with the measured $387 \pm 22\,\mathrm{km\,s^{-1}}$. The fixed $\beta_J = 11/50$ sensitivity gives $350.47\,\mathrm{km\,s^{-1}}$ for the stellar-only field and $401.34\,\mathrm{km\,s^{-1}}$ for CPTG. Both CPTG values lie within 1σ of the measured support without adding a dark-halo source to the local gravitational equation.

The same retained field then supplies a second, post-solution observable. Structural Mode is $N_{\mathrm{sph}} = 3.04935$ and $N_{\mathrm{obl}} = 3.06602$, placing XMM-VID1-2075 at the current Intermediate/Early CSMI transition. This does not reclassify the observed elliptical slow rotator as a spiral. It reports the radial organization of the CPTG field using the present CSMI catalog vocabulary.

The standard dark-matter formation framework must account for the early assembly, quenching, and angular-momentum loss of this extreme system inside its halo-plus-baryon history. CPTG makes a different gravitational claim: for the observed bound-system state, the measured stellar source plus the CPTG geometric response is sufficient to close the effective-radius pressure support and to produce an independent structural-mode readout without an unseen halo source in the local solve.

The present paper is therefore a positive native-phase, effective-radius Jeans, and Structural Mode validation under its declared assumptions. The next stronger test is a bin-by-bin reconstruction of the published JWST/NIRSpec kinematic field with an axisymmetric dynamical treatment and a correspondingly resolved Structural Mode source model.

Data, Code, and Evidence Availability

The observational inputs used in this study are drawn principally from the final JWST/NIRSpec and JWST/NIRCam analysis of Forrest et al. [1]. The source observations are publicly archived through the Mikulski Archive for Space Telescopes under the JWST program identified by the source paper. The source-distance convention is documented by Forrest et al. [2]; the conversion applied here is stated explicitly in Eq. (9). The CPTG quantities reported here are deterministic evaluations of the native-branch, distance-conversion, field, Jeans, and Structural Mode equations stated in the manuscript. Supporting CPTG code and research references are maintained at `https://github.com/CLG2025/CPTG`. Numerical implementations retain additional guard digits for reproducibility; manuscript values are rounded to the precision appropriate to interpretation.

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